Ex.no:3	# DATA CLEANING

Aim:

Algorithm:

Check Missing Values:

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check missing values using df.isnull().
6. Count missing values using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the results.

Drop Missing Values

1. Start.
2. Import the pandas library.
3. Load the dataset into a DataFrame (df).
4. Drop missing rows using dropna().
5. Drop rows (axis=0) or columns (axis=1).
6. Use how='any' to remove rows/columns containing any missing value.
7. Use how='all' to remove rows/columns only if all values are missing.
8. Use thresh=N to keep rows/columns with at least N non-missing values.
9. Use subset=['column_name'] to drop rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned DataFrame.

Program:

```
import pandas as pd
# Load dataset
df = pd.read_csv("indexData.csv")

print("===== DATASET INFO =====")
df.info()

print("\n===== MISSING VALUES IN EACH COLUMN =====")
print(df.isnull().sum())

print("\n===== TOTAL MISSING VALUES =====")
print(df.isnull().sum().sum())

print("\n===== ANY MISSING VALUES? =====")
print(df.isnull().values.any())

print("\n===== NON-MISSING VALUES COUNT =====")
print(df.notnull().sum())



import pandas as pd

df = pd.read_csv("hotel_bookings.csv")

# Fill all missing values with 0
df_constant = df.fillna(0)
print("===== CONSTANT VALUE =====")
print(df_constant.head())

# Fill 'children' column with mean
```

```
df_mean = df.copy()
df_mean["children"] = df_mean["children"].fillna(df_mean["children"].mean())
print("\n===== MEAN =====")
print(df_mean.head())

# Fill 'children' column with median
df_median = df.copy()
df_median["children"] = df_median["children"].fillna(df_median["children"].median())
print("\n===== MEDIAN =====")
print(df_median.head())

# Fill 'country' column with mode
df_mode = df.copy()
df_mode["country"] = df_mode["country"].fillna(df_mode["country"].mode()[0])
print("\n===== MODE =====")
print(df_mode.head())

# Forward Fill
df_ffill = df.ffill()
print("\n===== FORWARD FILL =====")
print(df_ffill.head())

# Backward Fill
df_bfill = df.bfill()
print("\n===== BACKWARD FILL =====")
print(df_bfill.head())
import pandas as pd

df = pd.read_csv("hotel_bookings.csv")

# Drop rows with missing values
df_drop_rows = df.dropna()
```

```
print("===== DROP ROWS =====")
print(df_drop_rows.head())

# Drop columns with missing values
df_drop_cols = df.dropna(axis=1)
print("\n===== DROP COLUMNS =====")
print(df_drop_cols.head())

# Drop rows having any missing value
df_any = df.dropna(how="any")
print("\n===== DROP HOW='ANY' =====")
print(df_any.head())

# Drop rows where all values are missing
df_all = df.dropna(how="all")
print("\n===== DROP HOW='ALL' =====")
print(df_all.head())

# Keep rows with at least 30 non-null values (32 columns total)
df_thresh = df.dropna(thresh=30)
print("\n===== THRESH=30 =====")
print(df_thresh.head())

# Drop rows where 'country' or 'children' is missing
df_subset = df.dropna(subset=["country", "children"])
print("\n===== SUBSET (country, children) =====")
print(df_subset.head())

# Drop missing values permanently
df.dropna(inplace=True)
print("\n===== AFTER inplace=True =====")
print(df.head())
```

Output:

```
----- DATASET INFO -----
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 112457 entries, 0 to 112456
Data columns (total 8 columns):
 #   Column     Non-Null Count   Dtype
---  ------     --------------   -----
 0   Index      112457 non-null  object
 1   Date       112457 non-null  object
 2   Open       110253 non-null  float64
 3   High       110253 non-null  float64
 4   Low        110253 non-null  float64
 5   Close      110253 non-null  float64
 6   Adj Close  110253 non-null  float64
 7   Volume     110253 non-null  float64
dtypes: float64(6), object(2)
memory usage: 6.9+ MB

----- MISSING VALUES IN EACH COLUMN -----
Index           0
Date            0
Open         2204
High         2204
Low          2204
Close        2204
Adj Close    2204
Volume       2204
dtype: int64

----- TOTAL MISSING VALUES -----
13224

----- ANY MISSING VALUES? -----
True

----- NON-MISSING VALUES COUNT -----
Index        112457
Date         112457
Open         110253
High         110253
Low          110253
Close        110253
Adj Close    110253
Volume       110253
dtype: int64

----- CONSTANT VALUE -----
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
----- MEAN -----
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== MEDIAN =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== MODE =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== FORWARD FILL =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== BACKWARD FILL =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== DROP ROWS =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== DROP COLUMNS =====
```

	Index	Date
0	NYA	1965-12-31
1	NYA	1966-01-03
2	NYA	1966-01-04
3	NYA	1966-01-05
4	NYA	1966-01-06

```
===== DROP HOW='ANY' =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== DROP HOW='ALL' =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== THRESH = 5 =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== SUBSET (Open, Close) =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

```
===== AFTER inplace=True =====
```

	Index	Date	Open	High	Low	Close	Adj Close	Volume
0	NYA	1965-12-31	528.690002	528.690002	528.690002	528.690002	528.690002	0.0
1	NYA	1966-01-03	527.210022	527.210022	527.210022	527.210022	527.210022	0.0
2	NYA	1966-01-04	527.840027	527.840027	527.840027	527.840027	527.840027	0.0
3	NYA	1966-01-05	531.119995	531.119995	531.119995	531.119995	531.119995	0.0
4	NYA	1966-01-06	532.070007	532.070007	532.070007	532.070007	532.070007	0.0

Result: